Research Question 2: Violent Crime by Reporting District in Los Angeles

2026-04-30

Research Question 2

Which areas in Los Angeles experience the highest rates of violent crime per district (Rpt Dist No), and how are these patterns distributed?

This analysis examines the geographic distribution of violent crime across Los Angeles reporting districts (`Rpt Dist No`).

Because population by reporting district is not available, the rate used in this analysis is defined as:

violent crimes per 1,000 total reported crimes within each reporting district

Load libraries

```
library(readr)
library(dplyr)
library(lubridate)
library(ggplot2)
library(knitr)
library(scales)
```

Load datasets

```
# Full dataset of all reported crimes
all_crime_df <- read_csv("Fixed_Date_version.csv")

# Dataset already filtered to violent crimes
violent_df <- read_csv("Violent_Crimes.csv")
```

Clean date columns

```
if (!inherits(all_crime_df$`DATE OCC`, "POSIXct") && !inherits(all_crime_df$`DATE OCC`, "Date")) {
  all_crime_df$`DATE OCC` <- parse_date_time(
    all_crime_df$`DATE OCC`,
    orders = c("Y b d I:M:S p", "Y-m-d H:M:S", "m/d/Y H:M", "m/d/Y I:M:S p")
  )
}

if (!inherits(violent_df$`DATE OCC`, "POSIXct") && !inherits(violent_df$`DATE OCC`, "Date")) {
  violent_df$`DATE OCC` <- parse_date_time(
    violent_df$`DATE OCC`,
    orders = c("Y b d I:M:S p", "Y-m-d H:M:S", "m/d/Y H:M", "m/d/Y I:M:S p")
```

```
  )
}
```

Analysis from 2020–2023

```
all_crime_df <- all_crime_df %>%
  mutate(year = year(`DATE OCC`)) %>%
  filter(year %in% 2020:2023)

violent_df <- violent_df %>%
  mutate(year = year(`DATE OCC`)) %>%
  filter(year %in% 2020:2023)
```

Compute district-level totals and rates

- total number of crimes
- total number of violent crimes
- violent crime rate per 1,000 crimes
- percent of crimes that are violent

```
district_total <- all_crime_df %>%
  group_by(`Rpt Dist No`, `AREA NAME`) %>%
  summarise(
    total_crime_count = n(),
    .groups = "drop"
  )

district_violent <- violent_df %>%
  group_by(`Rpt Dist No`, `AREA NAME`) %>%
  summarise(
    violent_count = n(),
    .groups = "drop"
  )

district_summary <- district_total %>%
  left_join(district_violent, by = c("Rpt Dist No", "AREA NAME")) %>%
  mutate(
    violent_count = ifelse(is.na(violent_count), 0, violent_count),
    violent_rate_per_1000 = (violent_count / total_crime_count) * 1000,
    violent_percent = (violent_count / total_crime_count) * 100
  ) %>%
  arrange(desc(violent_rate_per_1000))

kable(
  head(district_summary, 15),
  digits = 2,
  caption = "Top 15 Reporting Districts by Violent Crime Rate"
)
```

Table 1: Top 15 Reporting Districts by Violent Crime Rate

Rpt Dist No	AREA NAME	total_crime_count	violent_count	violent_rate_per_1000	violent_percent
225	Rampart	1	1	1000	100
242	Rampart	1	1	1000	100
252	Rampart	1	1	1000	100
716	Wilshire	2	2	1000	100
772	Wilshire	1	1	1000	100
800	West LA	1	1	1000	100
945	Van Nuys	1	1	1000	100
995	Van Nuys	1	1	1000	100
1000	West Valley	1	1	1000	100
1013	West Valley	1	1	1000	100
1014	West Valley	1	1	1000	100
1016	West Valley	1	1	1000	100
1042	West Valley	1	1	1000	100
1046	West Valley	1	1	1000	100
1048	West Valley	1	1	1000	100

Top districts

```r
top10_rate <- district_summary %>%
  slice_max(order_by = violent_rate_per_1000, n = 10)

top10_count <- district_summary %>%
  slice_max(order_by = violent_count, n = 10)

kable(top10_rate, digits = 2, caption = "Top 10 Districts by Violent Crime Rate")
```

Table 2: Top 10 Districts by Violent Crime Rate

Rpt Dist No	AREA NAME	total_crime_count	violent_count	violent_rate_per_1000	violent_percent
225	Rampart	1	1	1000	100
242	Rampart	1	1	1000	100
252	Rampart	1	1	1000	100
716	Wilshire	2	2	1000	100
772	Wilshire	1	1	1000	100
800	West LA	1	1	1000	100
945	Van Nuys	1	1	1000	100
995	Van Nuys	1	1	1000	100
1000	West Valley	1	1	1000	100
1013	West Valley	1	1	1000	100
1014	West Valley	1	1	1000	100
1016	West Valley	1	1	1000	100
1042	West Valley	1	1	1000	100
1046	West Valley	1	1	1000	100
1048	West Valley	1	1	1000	100
1087	West Valley	1	1	1000	100
1300	Newton	2	2	1000	100

Rpt Dist No	AREA NAME	total_crime_count	violent_count	violent_rate_per_1000	violent_percent
1537	N Hollywood	1	1	1000	100
1542	N Hollywood	2	2	1000	100
1558	N Hollywood	1	1	1000	100
1648	Foothill	1	1	1000	100
1686	Foothill	1	1	1000	100
1700	Devonshire	1	1	1000	100
1732	Devonshire	1	1	1000	100
1736	Devonshire	1	1	1000	100

```r
kable(top10_count, digits = 2, caption = "Top 10 Districts by Violent Crime Count")
```

Table 3: Top 10 Districts by Violent Crime Count

Rpt Dist No	AREA NAME	total_crime_count	violent_count	violent_rate_per_1000	violent_percent
645	Hollywood	4149	1493	359.85	35.98
646	Hollywood	3655	1454	397.81	39.78
636	Hollywood	3372	1324	392.65	39.26
1822	Southeast	2755	1305	473.68	47.37
1249	77th Street	2086	1226	587.73	58.77
1269	77th Street	2319	1163	501.51	50.15
1239	77th Street	2469	1153	466.99	46.70
1802	Southeast	2298	1151	500.87	50.09
119	Central	2429	1139	468.92	46.89
152	Central	2512	1114	443.47	44.35

Area-level summary

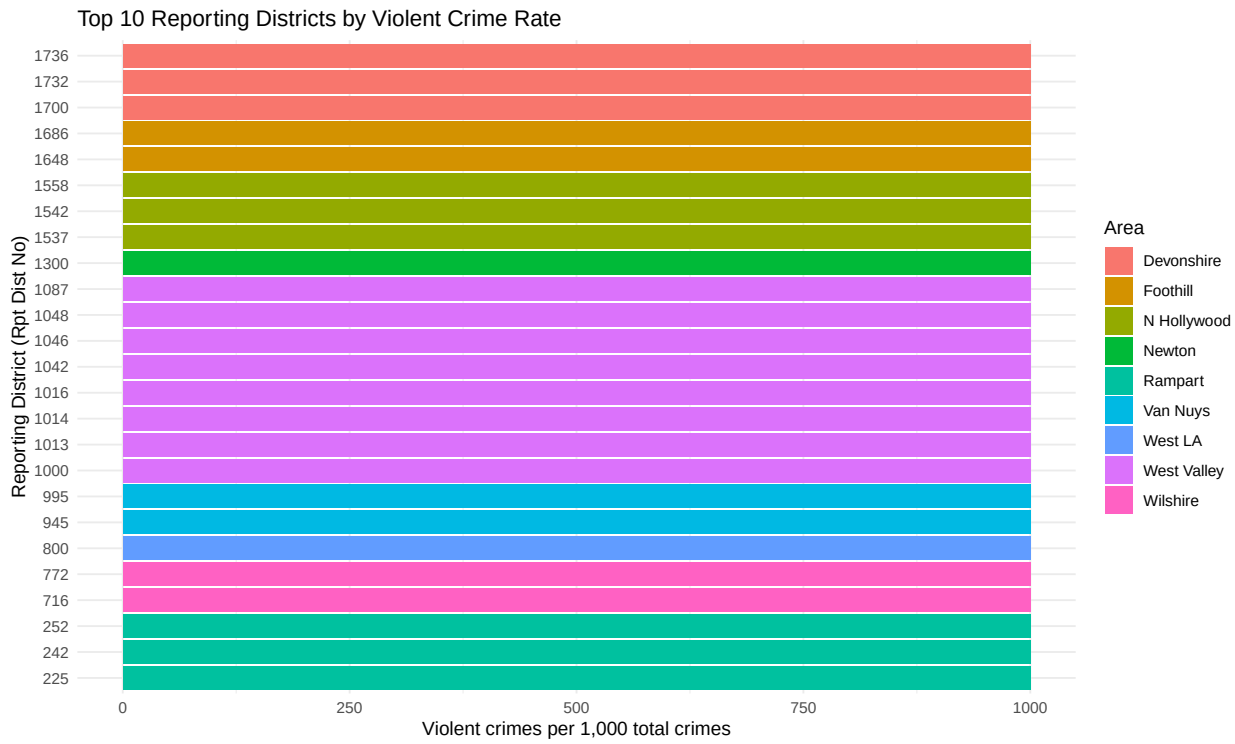
```r
area_summary <- district_summary %>%
  group_by(`AREA NAME`) %>%
  summarise(
    total_violent_crimes = sum(violent_count, na.rm = TRUE),
    total_crimes = sum(total_crime_count, na.rm = TRUE),
    avg_district_rate_per_1000 = mean(violent_rate_per_1000, na.rm = TRUE),
    median_district_rate_per_1000 = median(violent_rate_per_1000, na.rm = TRUE),
    district_count = n(),
    .groups = "drop"
  ) %>%
  mutate(
    area_violent_rate_per_1000 = (total_violent_crimes / total_crimes) * 1000
  ) %>%
  arrange(desc(area_violent_rate_per_1000))

kable(area_summary, digits = 2, caption = "Violent Crime Summary by LAPD Area")
```

Table 4: Violent Crime Summary by LAPD Area

AREA NAME	total_violent_crimes	total_crimes	avg_district_rate_per_1000	median_district_rate_per_1000	district_count	area_violent_rate_per_1000
77th Street	24224	52367	437.36	439.00	40	462.58
Southeast	19119	42212	441.56	452.47	34	452.93
Newton	16003	41912	410.07	409.81	45	381.82
Rampart	14937	39486	363.10	382.65	54	378.29
Southwest	16844	46862	352.17	357.16	58	359.44
Central	20048	56912	381.28	340.82	53	352.26
Hollenbeck	10922	31288	356.15	358.08	63	349.08
Olympic	14677	42573	344.77	358.51	53	344.75
Harbor	11816	34521	311.42	341.16	51	342.28
Hollywood	14569	44311	286.79	316.33	40	328.79
Mission	10786	33227	307.00	308.25	64	324.62
Foothill	8488	27804	299.03	292.78	67	305.28
West Valley	9999	35111	352.21	285.85	62	284.78
Topanga	9199	34113	262.61	253.97	53	269.66
Van Nuys	9187	35435	253.55	230.98	55	259.26
N Hollywood	10834	41927	289.38	260.33	63	258.40
Wilshire	10192	40167	280.29	262.09	64	253.74
Devonshire	8607	34202	269.32	243.25	74	251.65
Northeast	9025	36023	252.02	239.78	65	250.53
Pacific	10366	49298	228.34	198.06	75	210.27
West LA	7115	38626	197.50	179.68	70	184.20

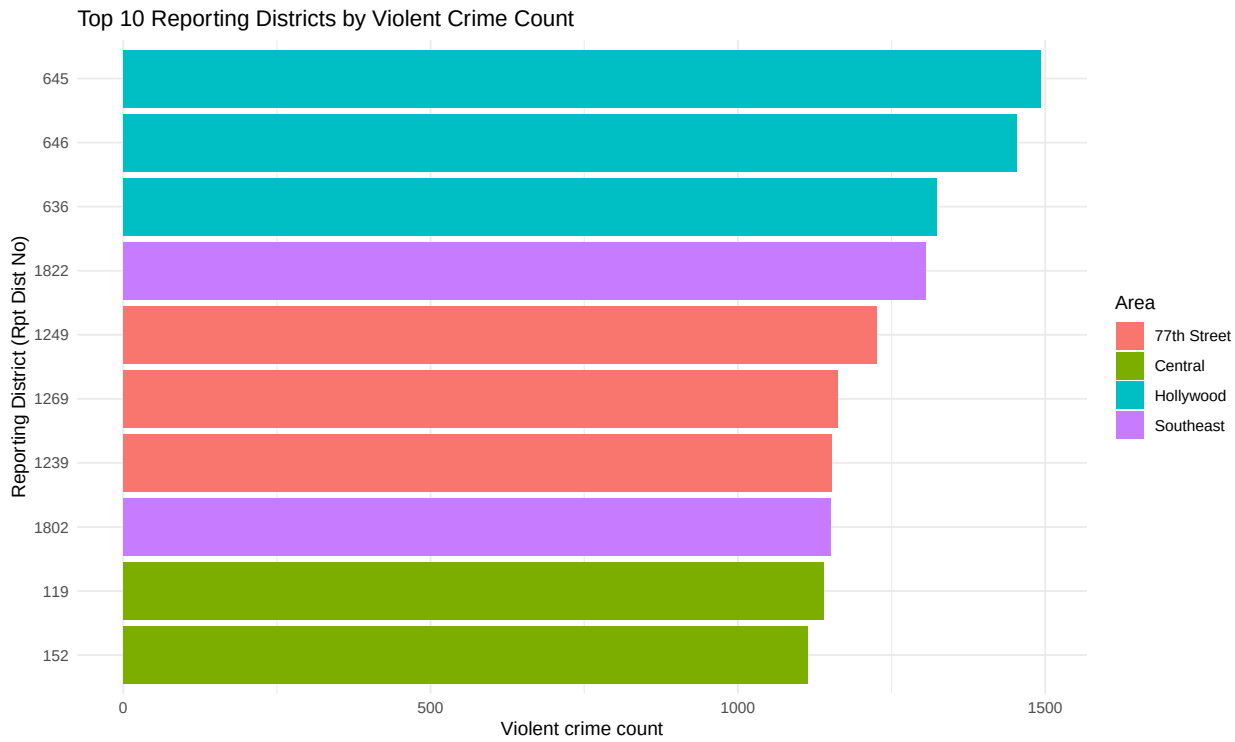
Visualization 1: Top 10 districts by violent crime rate

```
ggplot(
  top10_rate,
  aes(x = reorder(as.factor(`Rpt Dist No`), violent_rate_per_1000),
      y = violent_rate_per_1000,
      fill = `AREA NAME`)
) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top 10 Reporting Districts by Violent Crime Rate",
    x = "Reporting District (Rpt Dist No)",
    y = "Violent crimes per 1,000 total crimes",
    fill = "Area"
  ) +
  theme_minimal()
```



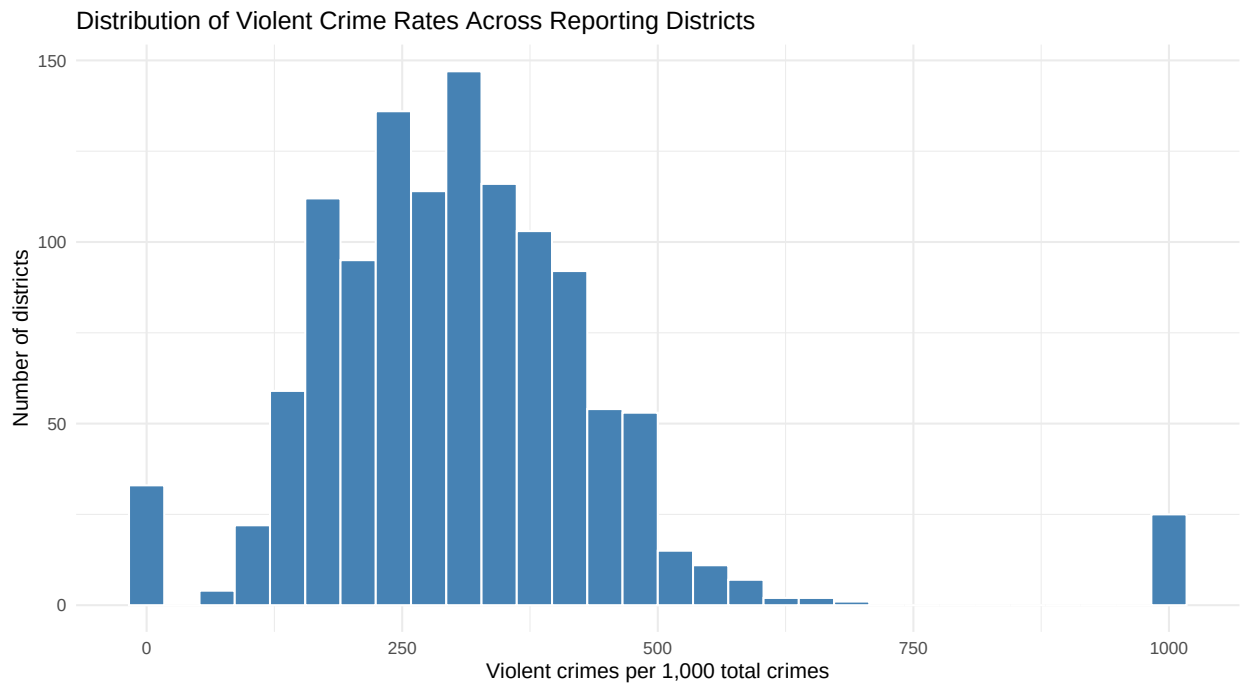
Visualization 2: Top 10 districts by violent crime count

```
ggplot(
  top10_count,
  aes(x = reorder(as.factor(`Rpt Dist No`), violent_count),
      y = violent_count,
      fill = `AREA NAME`)
) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top 10 Reporting Districts by Violent Crime Count",
    x = "Reporting District (Rpt Dist No)",
    y = "Violent crime count",
    fill = "Area"
  ) +
  theme_minimal()
```



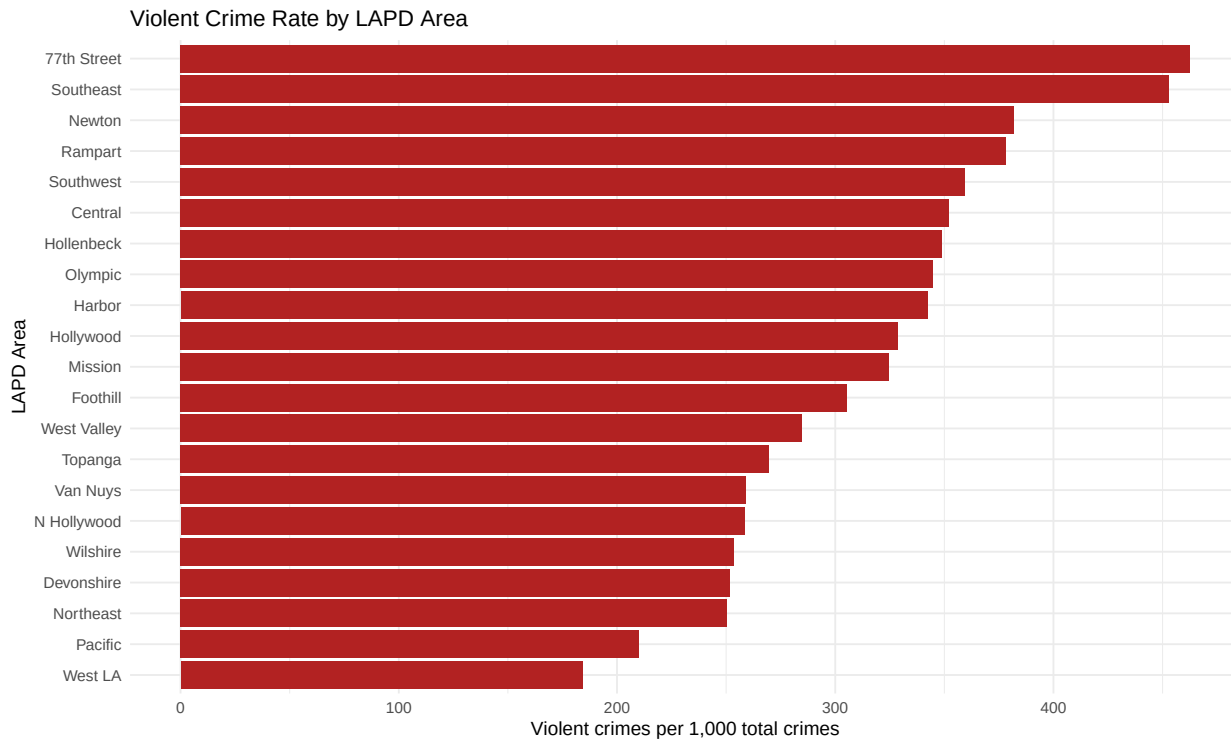
Visualization 3: Distribution of district violent crime rates

```
ggplot(district_summary, aes(x = violent_rate_per_1000)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  labs(
    title = "Distribution of Violent Crime Rates Across Reporting Districts",
    x = "Violent crimes per 1,000 total crimes",
    y = "Number of districts"
  ) +
  theme_minimal()
```



Visualization 4: Violent crime rate by area

```
ggplot(
  area_summary,
  aes(x = reorder(`AREA NAME`, area_violent_rate_per_1000),
      y = area_violent_rate_per_1000)
) +
  geom_col(fill = "firebrick") +
  coord_flip() +
  labs(
    title = "Violent Crime Rate by LAPD Area",
    x = "LAPD Area",
    y = "Violent crimes per 1,000 total crimes"
  ) +
  theme_minimal()
```



Spatial distribution

This map is an approximate spatial summary using the median latitude and longitude of violent incidents within each district.

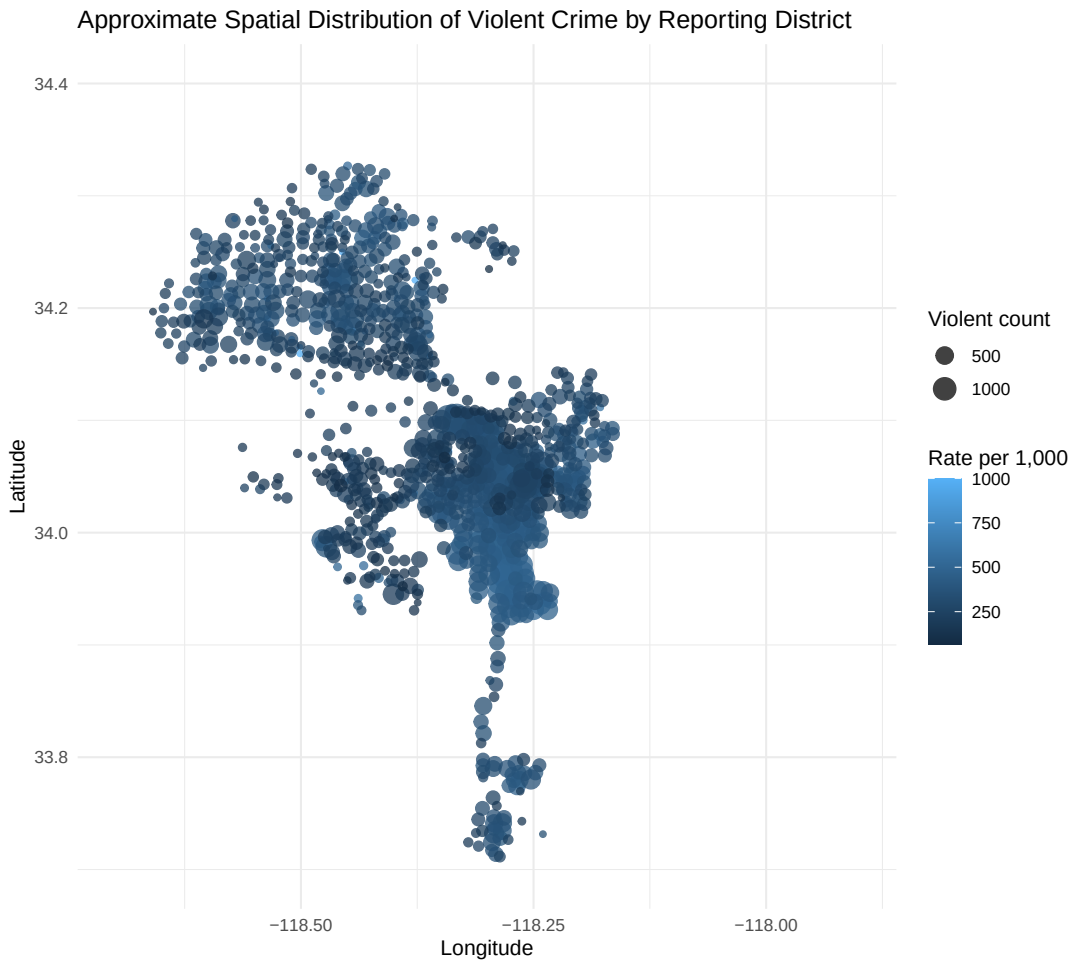
It is not a true district boundary map, but it helps show how high-violence districts are distributed geographically.

```
district_locations <- violent_df %>%
  filter(!is.na(LAT), !is.na(LON), LAT != 0, LON != 0) %>%
  group_by(`Rpt Dist No`) %>%
  summarise(
    median_lat = median(LAT, na.rm = TRUE),
    median_lon = median(LON, na.rm = TRUE),
    .groups = "drop"
  )

district_map <- district_summary %>%
  left_join(district_locations, by = "Rpt Dist No")
```

Visualization 5: Approximate district map

```
ggplot(
  district_map %>% filter(!is.na(median_lat), !is.na(median_lon)),
  aes(x = median_lon, y = median_lat,
      size = violent_count, color = violent_rate_per_1000)
) +
  geom_point(alpha = 0.75) +
  coord_quickmap(xlim = c(-118.7, -117.9), ylim = c(33.7, 34.4)) +
  labs(
```

```
    title = "Approximate Spatial Distribution of Violent Crime by Reporting District",
    x = "Longitude",
    y = "Latitude",
    size = "Violent count",
    color = "Rate per 1,000"
  ) +
  theme_minimal()
```



Key results

```
highest_rate_district <- district_summary %>% slice_max(violent_rate_per_1000, n = 1)
highest_count_district <- district_summary %>% slice_max(violent_count, n = 1)
highest_area <- area_summary %>% slice_max(area_violent_rate_per_1000, n = 1)

cat("District with the highest violent crime rate:\n")
```

```
## District with the highest violent crime rate:
```

```
print(highest_rate_district)
```

```
## # A tibble: 25 x 6
##    `Rpt Dist No` `AREA NAME` total_crime_count violent_count
##            <dbl> <chr>                   <int>         <dbl>
```

```
## 1           225 Rampart                     1             1
## 2           242 Rampart                     1             1
## 3           252 Rampart                     1             1
## 4           716 Wilshire                    2             2
## 5           772 Wilshire                    1             1
## 6           800 West LA                     1             1
## 7           945 Van Nuys                    1             1
## 8           995 Van Nuys                    1             1
## 9          1000 West Valley                 1             1
## 10         1013 West Valley                 1             1
## # i 15 more rows
## # i 2 more variables: violent_rate_per_1000 <dbl>, violent_percent <dbl>
```

```r
cat("\nDistrict with the highest violent crime count:\n")
```

```
##
## District with the highest violent crime count:
```

```r
print(highest_count_district)
```

```
## # A tibble: 1 x 6
##   `Rpt Dist No` `AREA NAME` total_crime_count violent_count
##           <dbl> <chr>                   <int>         <dbl>
## 1           645 Hollywood                4149          1493
## # i 2 more variables: violent_rate_per_1000 <dbl>, violent_percent <dbl>
```

```r
cat("\nArea with the highest violent crime rate:\n")
```

```
##
## Area with the highest violent crime rate:
```

```r
print(highest_area)
```

```
## # A tibble: 1 x 7
##   `AREA NAME` total_violent_crimes total_crimes avg_district_rate_per_1000
##   <chr>                      <dbl>        <int>                      <dbl>
## 1 77th Street                24224        52367                       437.
## # i 3 more variables: median_district_rate_per_1000 <dbl>,
## #   district_count <int>, area_violent_rate_per_1000 <dbl>
```

Results summary

The analysis shows that violent crime is not evenly distributed across Los Angeles reporting districts. Instead, violent crime is concentrated in a smaller number of districts and areas.

Districts with the highest violent crime rate are not always the same districts with the highest violent crime count. This means that some districts have a larger share of violent crime relative to all reported crime, even if their total violent incident count is not the highest overall.

At the area level, some LAPD areas contain multiple districts with relatively high violent-crime rates, indicating geographic clustering.

The bar charts and histogram show that district-level violent crime varies substantially across Los Angeles, while the spatial point map shows that these high-rate districts are concentrated in particular locations rather than spread evenly across the city.

Conclusion

Using violent crimes per 1,000 total crimes as the district-level rate measure, the results show that violent crime is spatially concentrated rather than evenly distributed. Some reporting districts consistently stand out as higher-risk districts, and some LAPD areas contain multiple high-rate districts.